higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

CONCRETE STRUCTURES
NQF LEVEL 3

(12020343)

11 March 2019 (X-Paper)
09:00–12:00

This question paper consists of 7 pages.

NCV

TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

NC(V)

QUESTION 1: THE THEORY AND USE OF REINFORCED CONCRETE

1.1 Make TWO drawings to explain the difference between a *cantilever beam* and a *cantilever wall*. Indicate where the tension and compression forces are and also include the tension steel reinforcement. Label the drawings. (18)

1.2 Make use of simple drawings to explain the change in a column under load:

1.2.1 If there is no reinforcement in the column (2)

1.2.2 If there is no stirrups in the reinforced column (3)

1.3 Name TWO factors that cause cracking and failing of concrete. (2)

[25]

QUESTION 2: REINFORCED CONCRETE IN CONSTRUCTION

2.1 Choose an item or a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–F) next to the question number (2.1.1–2.1.5) in the ANSWER BOOK.

COLUMN A		COLUMN B	
2.1.1	A process of increasing the strength of concrete by installing steel bars or wire mesh, strands of glass or reinforced plastic in the concrete or before casting the concrete	A	cold rolling
		B	reinforcement concrete
		C	high-yield steel
2.1.2	Steel is very brittle but has a high carbon content and high strength	D	mild steel
		E	cold working
2.1.3	Steel made of iron with less than 30% carbon and it can be easily cut, bent or weld	F	sheet-metal drawing
2.1.4	Changing the shape and size of a metal at room temperature		
2.1.5	Metalwork process in which metal is changed by passing it through rollers at a low temperature		

(5 × 1) (5)

2.2 Make a simple, neat drawing to explain the bar-drawing process. (4)

2.3 Name and explain FOUR different symbols used to designate different types of reinforced steel. (8)

2.4 Name THREE commonly used sizes/nominal diameter of reinforcement. (3)

2.5 Adding fibre to concrete greatly increases the toughness of the hardened concrete.

State FIVE factors to be considered when adding fibre to concrete. (5)
[25]

QUESTION 3: REINFORCING DOCUMENTATION

3.1 Explain the meaning of the following bar notation by means of a drawing:

4Y16-04 and 4Y16-05 EW 200 (4)

3.2 Interpret the following bar notation:

3.2.1 4Y20

3.2.2 5R8

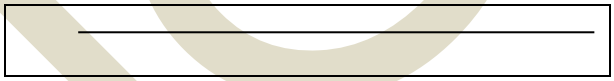
(2 × 3) (6)

3.3 Write down the meaning of the different types of lines used on the reinforcing drawing:

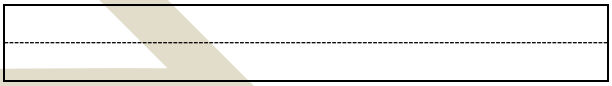
3.3.1



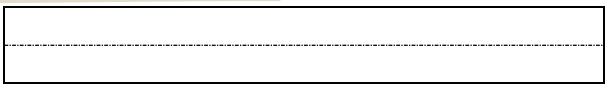
3.3.2



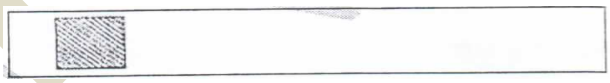
3.3.3



3.3.4

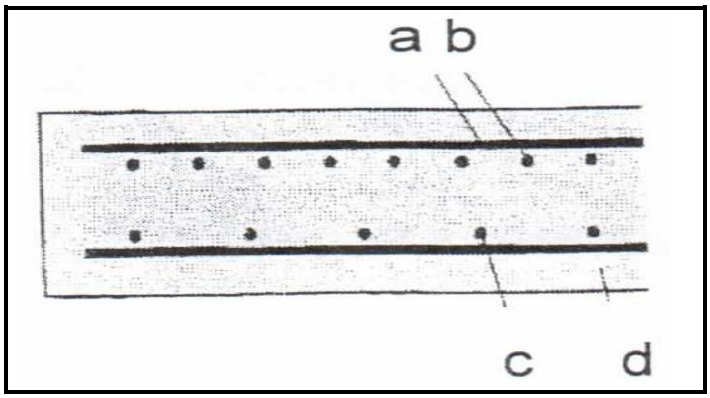


3.3.5



(5 × 1) (5)

3.4 Label the structure of the rebars indicated in the drawing below by writing only the answer next to the letter (a–d) in the ANSWER BOOK.



(4)

3.5 The following table shows cutting and bending tolerances. Complete the table by writing only the missing detail next to the question number (3.5.1–3.5.6) in the ANSWER BOOK.

METHOD OF FORMING	LENGTH	TOLERANCE
Cutting	3.5.1 …	3.5.2 …mm
Bending	Less than 3.5.3 …	3.5.4 … mm
	Between 1 m and 2 m	± 5 mm
	3.5.5 …	3.5.6 … mm

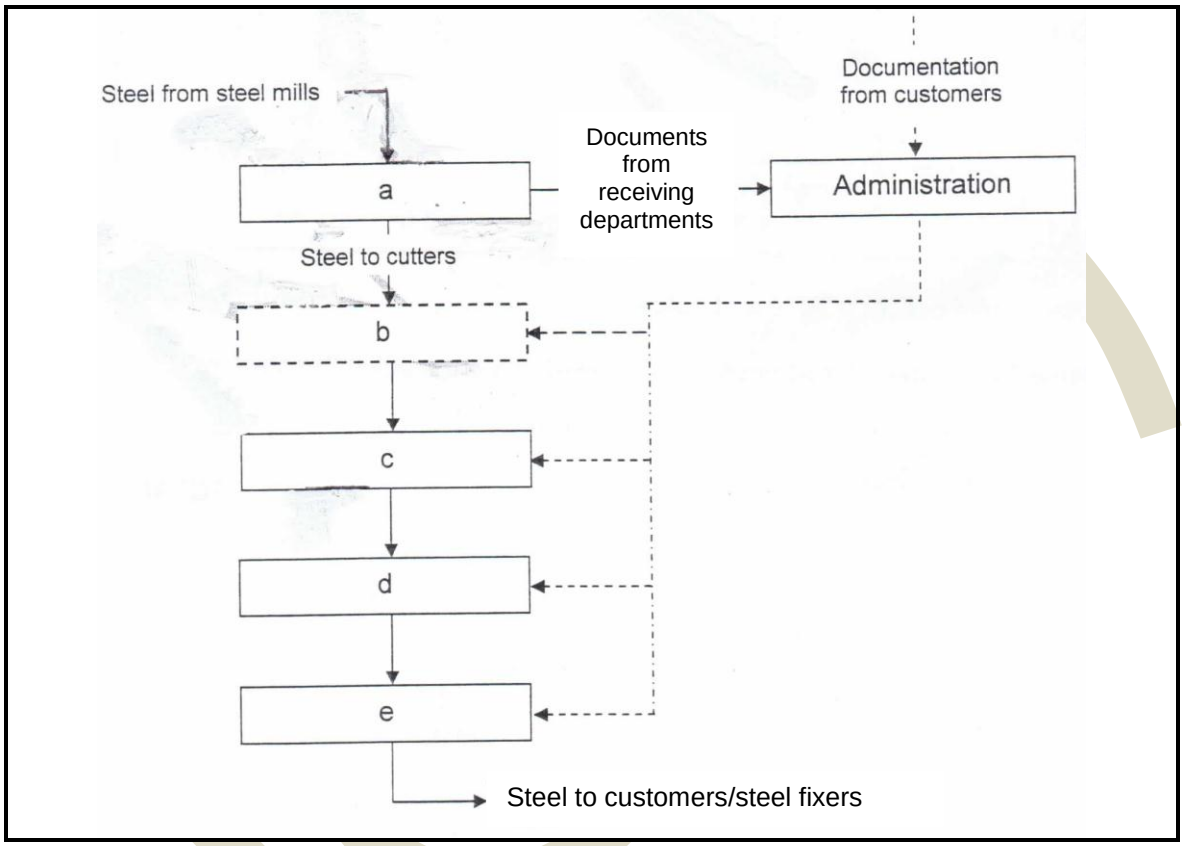
(6)
[25]

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QUESTION 4: FABRICATION OF REINFORCING STEEL

4.1 There are a variety of activities taking place in the steel yard.

Label the following process that takes place in a steel yard, by writing only the answer next to the letter (a-e) in the ANSWER BOOK.



(5)

4.2 Redraw the following table in the ANSWER BOOK and complete the table by giving an explanation of the information contained in the documents used in a steel yard as well as their purpose and the persons who use them.

TYPES OF DOCUMENTS	WHAT IS THEIR PURPOSE?	WHO USE THEM?	WHAT INFORMATION IS GIVEN?
Cutting and bending sheet	(2)	(1)	(1)
Bar label	(1)	(1)	(1)
Delivery note	(1)	(1)	(1)

(10)

4.3 Make a neat drawing of a bar label that will show the following information:

- Buyer is Mr Pitso
- Bar mark is 15
- Number of pieces, 20
- Shape, SABS code 43
- Cutting length is 1 200
- Grade and diameter of steel is R16 (10)

[25]

TOTAL: 100

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